INCH-POUND
MIL-A-24780(SH)
30 October 1992

MILITARY SPECIFICATION

ACTUATING ASSEMBLY, SINGLE LINEAR MOTION, FOR THE REMOTE ACTUATION OF QUICK ACTING VALVES, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a mechanical assembly used to apply a force through a single linear motion from a remote location for actuating quick acting valves. The assembly consists of an encased core mechanism, a valve end fitting, core junction boxes, and trip handles (see figure 1). Assemblies covered by this specification should be furnished in one class meeting the requirements specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 05Q42, Department of the Navy, 2531 National Center Bldg 3, Washington, DC 20362-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820
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SPECIFICATIONS

FEDERAL

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.
RR-W-410	- Wire Rope and Strand.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
DOD-G-24508	- Grease, High Performance, Multi-Purpose. (Metric)
MIL-G-46886	- Grease, Silicone.
MIL-C-81751	- Coating, Metallic-Ceramic.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-970	- Standards and Specifications, Order of Preference for the Selection of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 27 - Standard Specification for Steel Castings, Carbon, for General Application. (DoD adopted)
A 227 - Standard Specification for Steel Wire, Cold-Drawn for Mechanical Springs. (DoD adopted)
A 228 - Standard Specification for Steel Wire, Music Spring Quality. (DoD adopted)
A 229 - Standard Specification for Steel Wire, Oil-Tempered for Mechanical Springs. (DoD adopted)
A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)
A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes. (DoD adopted)
A 494 - Standard Specification for Castings, Nickel and Nickel Alloy.
A 527 - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process, Lock-Forming Quality. (DoD adopted)
A 568 - Standard Specification for Steel, Sheet, Carbon, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for. (DoD adopted)
B 16 - Standard Specification for Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines. (DoD adopted)
B 21 - Standard Specification for Naval Brass Rod, Bar, and Shapes. (DoD adopted)
B 26 - Standard Specification for Aluminum-Alloy Sand Castings. (DoD adopted)
B 121 - Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar. (DoD adopted)
B 633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)
D 1457 - Standard Specification for Polytetrafluoroethylene (PTFE) Molding and Extrusion Materials. (DoD adopted)
D 2116 - Standard Specification for FEP-Fluorocarbon Molding and Extrusion Materials. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)
D 4066 - Standard Specification for Nylon Injection and Extrusion Materials (PA). (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN PETROLEUM INSTITUTE (API)

STD 607 - Fire Test for Soft-Seated Quarter-Turn Valves. (DoD adopted)

(Application for copies should be addressed to the American Petroleum Institute, 1220 L Street, NW, Washington, DC 20005.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Reliability. The assemblies shall be designed and constructed with reliability as a paramount consideration. The suitability of the assembly shall be based on breaking strength, endurance, and ability to withstand the environment. Verification of suitability shall be achieved by successfully meeting the requirements in 3.2.1 through 3.2.3.

3.2.1 Breaking strength. The assembly components which transmit the valve actuation force shall have a breaking strength of at least five times the working load (see 4.7.1).

3.2.2 Endurance. The assemblies shall operate for 10,000 cycles (see 4.7.2).

3.2.3 Environment. The assemblies shall operate in a marine environment under the following ambient conditions:

(a) Temperature range of minus 40 to plus 140 degrees Fahrenheit (°F).
(b) Pressure range of 28 to 32 inches of mercury (Hg) absolute.
(c) Relative humidity of 0 to 100 percent.

The assemblies furnished shall be subjected to a temperature soak test (see 4.7.3).

3.3 Materials. The assembly materials shall be as specified in table I (see 6.7). Materials not specified shall be suitable for the intended use and may be selected in accordance with MIL-STD-970.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.3.2 Corrosion protection. Steel components shall be protected from the environment. Steel components which must be exposed to the environment shall be corrosion resistant steel or plated with chrome or zinc. Chrome plating shall be in accordance with QQ-C-320. Zinc plating shall be in accordance with ASTM A 527 or ASTM B 633. Nonmoving components may be protected by a metallic-ceramic coating in accordance with MIL-C-81751.

TABLE I. Assembly materials.

Name of part	Material specification
Core	QQ-N-281 RR-W-410 ASTM A 227 ASTM A 276
Casing	QQ-N-281 ASTM A 228 ASTM A 229 ASTM A 269 ASTM A 276 ASTM D 1457 ASTM D 2116 ASTM D 4066
Housings	QQ-N-281 ASTM A 494 QQ-C-390 QQ-S-763 ASTM A 27 ASTM A 276 ASTM A 568 ASTM B 121
Handles, fittings, and couplings	QQ-N-281 ASTM A 494 QQ-C-390 QQ-S-763 ASTM A 276 ASTM B 16 ASTM B 21 ASTM B 26

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3.4 Standard commercial product. Assemblies furnished may be the manufacturer's standard commercial product provided the requirements specified herein are met.

3.5 Design and construction. The basic assembly shall consist of a movable (single linear motion) core coupled at one end to a trip handle. The other end of the core shall be coupled to the valve (during installation). Actuation of the trip handle shall transmit, via the core, the force and displacement necessary to trip the valve. The trip handle housing and core casing shall be suitable for mounting the assembly to ship structures (during installation). The assembly furnished shall contain the components specified in 3.7 and meet the requirements specified in 3.5.1 through 3.5.6 (see 6.3, appendix A, and appendix B).

3.5.1 Trip handle safety lock. Trip handles shall contain a feature to prevent inadvertent actuation by personnel. The time required to override this feature shall not exceed 3 seconds (see 4.7.4).

3.5.2 Trip handle and core return. The assembly shall contain a spring or mechanism which returns the actuating members (core and trip handles) to their stowed position after the assembly has been actuated and the applied force removed.

3.5.3 Trip handle actuation motion. Trip handles shall be tee handles or levers as specified (see 6.2). Unless otherwise specified (see 6.2), the displacement required at a trip handle by the operator's hand to stroke the core and actuate the valve shall not exceed 10 inches for tee handles nor an arc length of 20 inches for levers.

3.5.4 Shock features. The assembly shall be constructed with a shock compensation tube allowing 4 inches of cable movement before engaging the trip mechanism to preclude inadvertent valve trip induced by shock loads (see 3.6.1 and 4.7.5). The following conditions which may cause inadvertent valve actuation shall be considered in the construction of the assembly.

(a) A relative displacement as large as 4 inches may occur between the valve and the ship structures.
(b) Any mass (for example, a valve end fitting) coupled to the valve actuating mechanism will increase inertial loads and reduce the shock resistance of the valve.
(c) Unauthorized adjustments by personnel can reduce the shock resistance of the system.

3.5.5 Assembly couplings. Couplings shall be provided between the assembly components. Couplings shall be constructed to permit assembly and disassembly with standard tools.

3.5.6 Lubrication and seals. The assembly shall not require lubrication or it shall be permanently lubricated during manufacture. Permanently lubricated assemblies shall contain seals to retain lubricant and prevent entry of contaminants (see 4.7.2). Lubricants shall meet the requirements of MIL-G-46886 or DOD-G-24508.

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3.6 Performance. The assembly shall meet the performance requirements specified in 3.6.1 through 3.6.7.

3.6.1 Mechanical shock. The assembly shall meet the high-impact shock requirements of MIL-S-901 for grade A, class I equipment (see 4.7.5). The equipment classification shall be hull mounted.

3.6.2 Vibration. The assembly shall meet the requirements of MIL-STD-167-1 for type 1 equipment (see 4.7.6).

3.6.3 Working configuration. The working configuration for the assemblies furnished may vary (see 6.2). Unless otherwise specified (see 6.2), directional changes shall be accomplished by a 90 degree bend at a radius of 12 inches.

3.6.4 Valve parameters. The valves to be actuated by the assemblies furnished may vary in design. The valve parameters which interface with the assemblies furnished shall be as specified (see 6.2).

3.6.5 Actuation force. Unless otherwise specified (see 6.2), the force required at a trip handle to actuate the valve when the assembly is in the configuration specified (see 3.6.3) shall not exceed 50 pounds (see 4.7.7).

3.6.6 Orientation. The assembly shall function when placed in any orientation (see 4.7.8).

3.6.7 Fire condition. The assembly shall function during exposure to fire (see 4.7.9).

3.7 Components. The components furnished shall meet the requirements specified in 3.7.1 through 3.7.5 (see figure 1 for component identification).

3.7.1 Runs. The term "runs" as used herein refers to a subassembly consisting of a flexible, linear motion, inner member (the core) which may consist of, but is not limited to, a solid or stranded construction in a flexible, metallic, outer member (the casing) of solid, stranded, or interlocked construction. The casing shall be suitable for mounting the runs to the ship structure. The length of each run and the two interfacing components (that is, valve, junction box, remote trip handle, or intermediate trip handle) shall be as specified (see 6.2). Runs shall be provided with a valve end fitting or couplings as specified (see 6.2).

3.7.2 Remote trip handle. Remote trip handles are located at the ends of the assembly and are coupled to a single run. Each remote trip handle shall be provided with a housing. The housing shall contain provisions for mounting to the ship structure.

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3.7.3 Intermediate trip handle. Intermediate trip handles may be in the assembly. They are located between a remote trip handle and the valve. Runs are coupled to both sides. Each intermediate trip handle shall be provided with a housing. The housing shall contain provisions for mounting to the ship structure. Internal to the housing shall be a device that transmits an actuation force (and core motion) generated from the remote trip handle or the intermediate trip handle.

3.7.4 Junction box. Junction boxes may be used in the assembly. Junction boxes shall contain features to couple two or more runs at one end to single run (or valve) at the other end. All junction boxes shall transmit forces and core motion from the input run to the output run (or valve). Junction boxes shall have provisions for mounting to the ship structure. The number of runs to be connected at each junction box and the type of component coupling (that is, valve or run) to be furnished at the output shall be as specified (see 6.2).

3.7.5 Couplings and miscellaneous parts. Couplings and miscellaneous parts necessary for installing the assembly shall be furnished with the assembly.

3.8 Identification. Each component of the assembly shall be permanently and legibly marked with the following information:

(a) Manufacturer's part or drawing number.
(b) Manufacturer's name or trademark.
(c) Manufacturer's serial number.

3.9 Workmanship. The finished product shall be free from burrs, slivers, surface scratches, sharp corners or edges, and paint runs or smears.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the visual and dimensional inspections specified in 4.5 and the tests specified in 4.7, in that order. The assembly must pass all inspections and tests specified to be considered acceptable.

4.3.1 Test sample. The first article test sample shall be representative of the assemblies furnished. Unless otherwise specified (see 6.5), it shall consist of two remote trip handles, one junction box, one intermediate trip handle, and one valve end fitting. The components shall be connected by runs, as shown on figure 1, so that a functional system is provided. The length of each run in the test sample shall not exceed the minimum amount necessary to complete the tests specified in 4.7.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual and dimensional inspections specified in 4.5 and the design data inspections specified in 4.6. Quality conformance inspection shall be accomplished on 100 percent of the assemblies offered. Only those assemblies which pass all inspections shall be offered for delivery.

4.5 Visual and dimensional inspection. The visual and dimensional inspection shall consist of the inspections listed in table II.

TABLE II. Visual and dimensional inspections.

Inspection	Requirement
Presence and operation of trip handle lock.	3.5.1
Presence of return mechanism.	3.5.2
Type and magnitude of trip handle motion.	3.5.3
Use of standard tools.	3.5.5
Number of components.	3.7
Length of each run.	3.7.1
Configuration of each junction box.	3.7.4
Proper identification.	3.8
Quality of workmanship.	3.9

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4.6 Design data inspections. Design data inspections shall be accomplished by engineering review of the documentation provided (see 6.3 and appendix B).

4.7 Test methods. The required tests are listed in table III (see 6.2 and 6.3). The tests shall be conducted on the test sample specified in 4.3.1. The tests shall be conducted in the following order: vibration, mechanical shock, temperature soak, lock release, orientation, endurance, actuation force, breaking load, and fire test.

TABLE III. Tests.

Inspection	Requirement	Test method
Breaking strength	3.2.1	4.7.1
Endurance	3.2.2, 3.5.6	4.7.2
Temperature soak	3.2.3	4.7.3
Lock release	3.5.1	4.7.4
Mechanical shock	3.5.4, 3.6.1	4.7.5
Vibration	3.6.2	4.7.6
Actuation force	3.6.5	4.7.7
Orientation	3.6.6	4.7.8
Fire	3.6.7	4.7.9

4.7.1 Breaking strength test. The test sample shall be tested with a 150 pound force applied at each remote trip handle. This shall be accomplished by applying the necessary resistive load to the valve end fitting. The configuration of the test sample during the test shall be as specified in 4.7.2. The assembly shall have passed the test if the test sample remains normally functional.

4.7.2 Endurance test. The test sample shall be cycled for 10,000 cycles by a force of 50 pounds applied at a remote trip handle. This shall be accomplished by applying the necessary resistive load to the valve end fitting. The configuration of the test sample during the test shall be representative of the least efficient configuration specified (see 3.6.3 and 6.2). The assembly shall have passed the test if, at the completion of the test, there are no visible signs of lubricant leakage and the test sample functions normally.

4.7.3 Temperature soak test. The test sample shall be subjected to an air atmosphere of 170°F and ambient pressure for 170 hours. The assembly shall have passed the test if the test sample remains normally functional.

4.7.3.1 Low temperature. No test is required for low temperature functionality.

4.7.4 Safety lock release test. A demonstration shall be conducted to verify that the design of the trip handle safety lock meets the 3-second release requirement.

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4.7.5 Mechanical shock test. This test shall be conducted on a floating shock platform (barge) in accordance with MIL-S-901. The test sample and a valve to be specified (see 6.2), shall be mounted to simulate a shipboard installation. The valve end fitting of the test sample shall be coupled to the valve to form a functional system. The assembly shall have passed the test if the valve does not trip and the test sample remains normally functional.

4.7.6 Vibration test. This test shall be conducted in accordance with MIL-STD-167-1. The assembly shall have passed the test if it functions normally following the test.

4.7.7 Actuation force test. The test sample shall be configured as specified in 4.7.2 and a resistive load equivalent to the maximum specified valve trip force (see 3.6.4) shall be applied at the valve end fitting. The assembly shall have passed the test if the actuation forces do not exceed the value specified in 3.6.5.

4.7.8 Orientation test. The actuation force test shall be repeated with the trip handles of the test sample oriented in the vertical, horizontal, and inverted positions.

4.7.9 Fire test. The test sample shall be subjected to flame in accordance with section 1.4 of API STD 607 as modified by additional requirements of this paragraph. The test sample shall be cycled continuously at a temperature of 1400 to 1800°F for 30 minutes by a force of 50 pounds applied at a remote trip handle. Force versus time for the duration of the test shall be recorded and the time interval between cycles noted. The flame shall envelope all components identified on figure 1 from the junction box to the valve end fitting, inclusive. The assembly shall have passed the test if it functions normally during the test.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Items shall be preserved in accordance with method IC-2 of MIL-P-116.

5.1.2 Level C. Items shall be cleaned, dried, and preserved in a manner that will afford protection against corrosion, deterioration, and mechanical or physical damage during shipment from the supply source to the first receiving activity. The requirements of ASTM D 3951 shall be met.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Items shall be packed in wood boxes conforming to PPP-B-621, class 2, style optional. When the gross weight exceeds 250 pounds, the shipping container shall have skids applied in accordance with applicable box specifications.

5.2.2 Level B. Items shall be packed in weather-resistant class fiberboard boxes conforming to PPP-B-636, or in wood boxes conforming to PPP-B-621, class 1, style optional. When the gross weight exceeds 250 pounds, the shipping container shall have skids applied in accordance with applicable box specifications.

5.2.3 Level C. Items shall be overpacked in such a manner which affords protection against damage during direct shipment from the supply source to the first receiving activity. The requirements of ASTM D 3951 shall be met.

5.3 Marking. In addition to any special marking required (see 6.2), unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The actuating assemblies covered by this specification are intended for use in applying a force through a linear motion from a remote location. The assemblies are intended for, but not limited to, actuating quick acting valves in shipboard applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Whether first article inspection is required (see 3.1).
(d) Type of trip handles to be furnished and the maximum actuation stroke for trip handles if different from the value specified (see 3.5.3).
(e) Working configuration data for each assembly (see 3.6.3):

(1) Quantity of remote and intermediate trip handles required.
(2) The following information for each trip handle:

a. Quantity of intermediate trip handles and junction boxes to be installed between the trip handle and the valve.
b. Total length of assembly and total degrees of bend to be installed between each trip handle and the valve.

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(3) Minimum bend radius if less than 12 inches.

(f) Parameters for the valve to be actuated (see 3.6.4):
(1) Length of core stroke required at the valve to actuate the valve.
(2) Force required at the valve to trip the valve.

(g) Maximum permissible actuation force if different from the amount specified (see 3.6.5).
(h) Information for each run ordered (see 3.7.1):

(1) Length.
(2) Type of components to be coupled to each end of the run.

(i) Information for each junction box (see 3.7.4):

(1) Number of runs to be coupled at each end of the junction box.
(2) If the junction box is to be coupled to the valve.

(j) Whether test plan approval is required (see 4.7).
(k) Least efficient assembly configuration (see 4.7.2).
(l) Details of valve to be used for shock test (see 4.7.5).
(m) Preservation and packing levels, and any special marking required (see 5.1, 5.2, and 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5 and appendix A	DI-DRPR-80651	Engineering drawings	-----
3.5, 4.6, and appendix B	DI-GDRQ-80650	Design data and calculations	-----
4.7	UDI-T-23881	Plans, test	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.8 Subject term (key word) listing.

Castings
Handles, trip
Mechanism, encased
Resistance, corrosion
Resistance, high-impact

Preparing activity:
Navy - SH
(Project 4820-N075)

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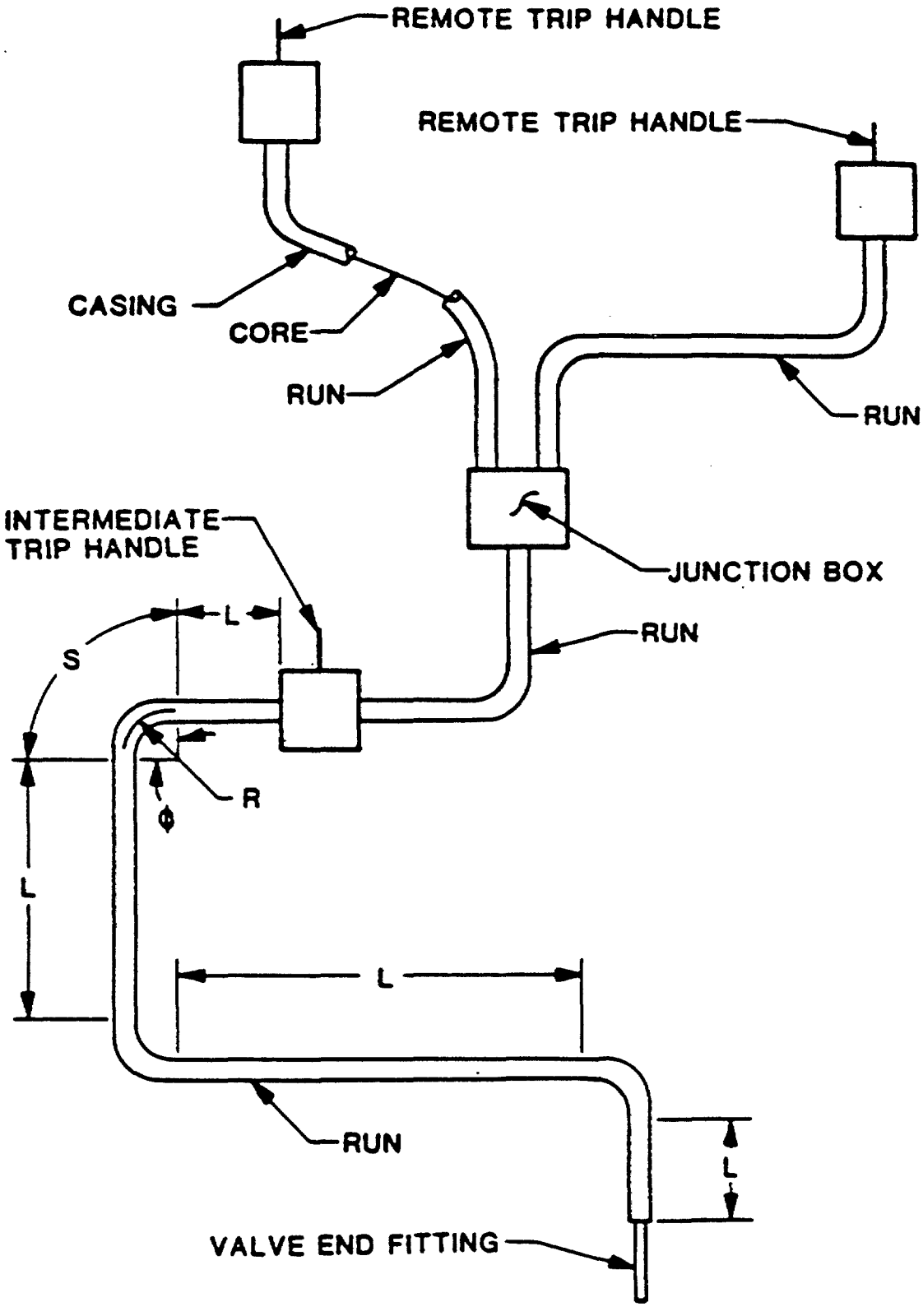


FIGURE 1. Mechanical actuating assembly nomenclature and component identification.

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APPENDIX A

ENGINEERING DRAWINGS
TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix details the technical content requirements to be furnished on or with the drawings. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENT

30.1 Drawing content. The following shall be furnished on or with the drawings:

(a) Sectional and assembly views of components.
(b) Outline dimensions.
(c) Dimensional location and size of mounting holes.
(d) Bill of material (indicating specification, grade, and condition of materials).
(e) General notes stating performance data including: minimum radius, load capacity, core stroke, and performance curves. Performance curves shall show efficiency in relation to length and total degrees of curvature for the assembly.

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APPENDIX B

DESIGN DATA AND CALCULATIONS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix details the technical content requirements for the actuation force test and other design data inspections as listed in table IV. This appendix is mandatory only when data item description DI-GDRQ-80650 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TECHNICAL CONTENT

30.1 Actuating force. The contractor shall furnish engineering calculations or test result which provide evidence that the actuating force does not exceed the amount specified (50 pounds as specified in 3.6.5 herein).

30.2 Design data inspections. The design data inspections may be accomplished by an engineering review of the documentation provided that should show that the requirements listed in table IV have been met.

TABLE IV. Design data inspections.

Inspection	Requirement
Breaking strength	3.2.1
Endurance	3.2.2
Actuation force	3.6.5

30.3 Submission of one data package. When more than one assembly is ordered, with identical acquisition requirements, submission of one data package for the total quantity will be sufficient.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-24780(SH)	30 OCTOBER 1992

3. DOCUMENT TITLE ACTUATING ASSEMBLY, SINGLE LINEAR MOTION FOR THE REMOTE ACTUARION OF QUICK ACTING VALVES, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL POINT OF CONTRACT: Oscar Ted Halverson/SEA 05Y23	(1) Commercial (703) 602-0367	(2) AUTOVON (AV) 332-0367
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND SEA 05Q42 2531 NATIONAL CENTER 3 WASHINGTON, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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